

Matrix functions and stability

A pointwise Lipschitz lower bound for the joint spectral radius

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Challenging because the sharp one-sided perturbation rate must survive reducibility; community impact is reliable lower stability estimates under data perturbation.

Context and notation

Let $\mathcal{H}_d$ denote the nonempty compact subsets of $\mathbb{C}^{d \times d}$. Use the spectral norm and its Hausdorff distance

$$d_H(\mathcal{M}, \mathcal{N}) = \max \left\{ \sup_{A \in \mathcal{M}} \inf_{B \in \mathcal{N}} \|A - B\|_2, \sup_{B \in \mathcal{N}} \inf_{A \in \mathcal{M}} \|A - B\|_2 \right\}.$$

The joint spectral radius is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

These definitions also apply to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

Problem statement

For every $d \geq 1$ and $\mathcal{M} \in \mathcal{H}_d$, do there exist $r, C > 0$ such that

$$\hat{\rho}(\mathcal{N}) \geq \hat{\rho}(\mathcal{M}) - Cd_H(\mathcal{M}, \mathcal{N}) \quad \text{if } d_H(\mathcal{M}, \mathcal{N}) < r?$$

Reference and status evidence

Epperlein and Wirth, [The joint spectral radius is pointwise Hölder continuous](#), §2, Conjecture 3 (P2). The conjectured exponent is one.

Scope

The reference family is fixed while the perturbed family varies. An answer controls how rapidly a stability diagnostic can decrease under data error; [MF-05](#) instead requires a two-sided estimate uniform over two varying families.

Additional status evidence

Searches combining “joint spectral radius” with “local Hölder”, “Lipschitz lower”, “trajectory bounds”, the authors’ names, and 2025/2026 found no later resolution. These searches supplement the explicit 2025 conjectures; they do not establish exhaustiveness. The three entries are separately named assertions in the source, not a count of dimensional cases.

Audit — 2026-09-10

Rechecked [Epperlein–Wirth](#), §1 and [Conjecture 3 \(P2\)](#). Local Lipschitz continuity gives the asserted lower bound at irreducible families. The general reducible case remains posed, and searches for Lipschitz lower bounds and later author work found no resolution.